

COFFEE SHOP ANALYTICS

From Raw Data to Business Strategy

CSV Data → PostgreSQL → Power BI — A Complete Pipeline

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- ✦ Role: Data Analyst Portfolio Project
- ✦ Date: June 2026

The Story Behind This Project

The Business Problem:

Every coffee shop owner faces a common challenge:

"How do we increase revenue without losing customers?"

The client wanted answers to three critical questions:

- ? 1. Should we increase prices? — And on which products?
- ? 2. Who are our most valuable customers? — And how do we keep them?
- ? 3. Which product categories are worth investing in? — And which should we phase out?

The Data:

- 450+ customers · 50 products (5 categories) · ~2,000 orders
- 3 years of data (2023–2025) · 3 CSV files → PostgreSQL → Power BI

Key Business Insights & Actions

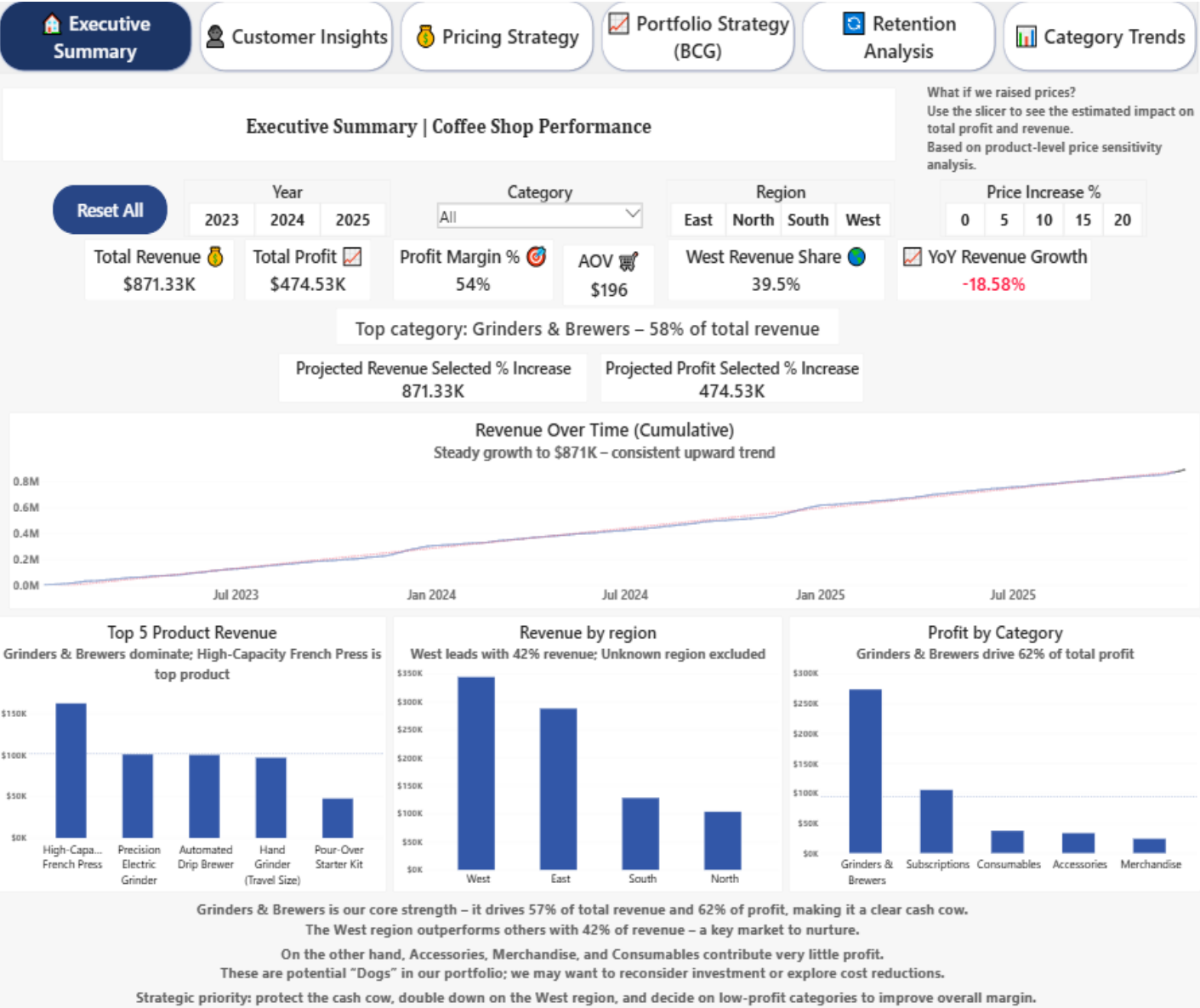
Here's what the data actually told me, and what I would do about it:

Insight	What Data Says	Business Action
Pricing Power	10 % increase shows positive profit impact.	Implement in phases. Start with Grinders & Brewers.
Customer Concentration	Top 10 customers = 11 % revenue (\$817K).	Launch loyalty program & premium subscriptions for top spenders.
Cash Cow vs. Dogs	Grinders & Brewers = 57% revenue , 62% profit (Cash Cow). Accessories, Merchandise, Consumables = Dogs.	Protect Cash Cow. Re-evaluate Dogs for discontinuation.
Retention Dip	100% → ~45% in second month.	Focus re-engagement campaigns on months 2-3.

Dashboard Overview — 6 Interactive Pages

Page	Focus	Key Features
1. Executive Summary	Overall business health	KPI cards, forecast, top 5 products
2. Customer Insights	Customer segmentation	RFM table, heatmap, Decomposition tree
3. Pricing Strategy	Price increase simulation	KPI cards, product table, scatter plot
4. Portfolio (BCG)	BCG Matrix strategy	BCG scatter plot, category table
5. Retention Analysis	Customer loyalty analysis	Cohort matrix, line chart
6. Category Trends	YoY revenue trends	Line chart, table with data bars

Executive Summary — The 30,000-foot View



Key Metrics:

- Total Revenue: \$871.33K
- Total Profit: \$474.53K
- Profit Margin: 54%
- AOV: \$196
- West Revenue Share: 42%
- YoY Revenue Growth: -18.58%

Revenue Over Time:

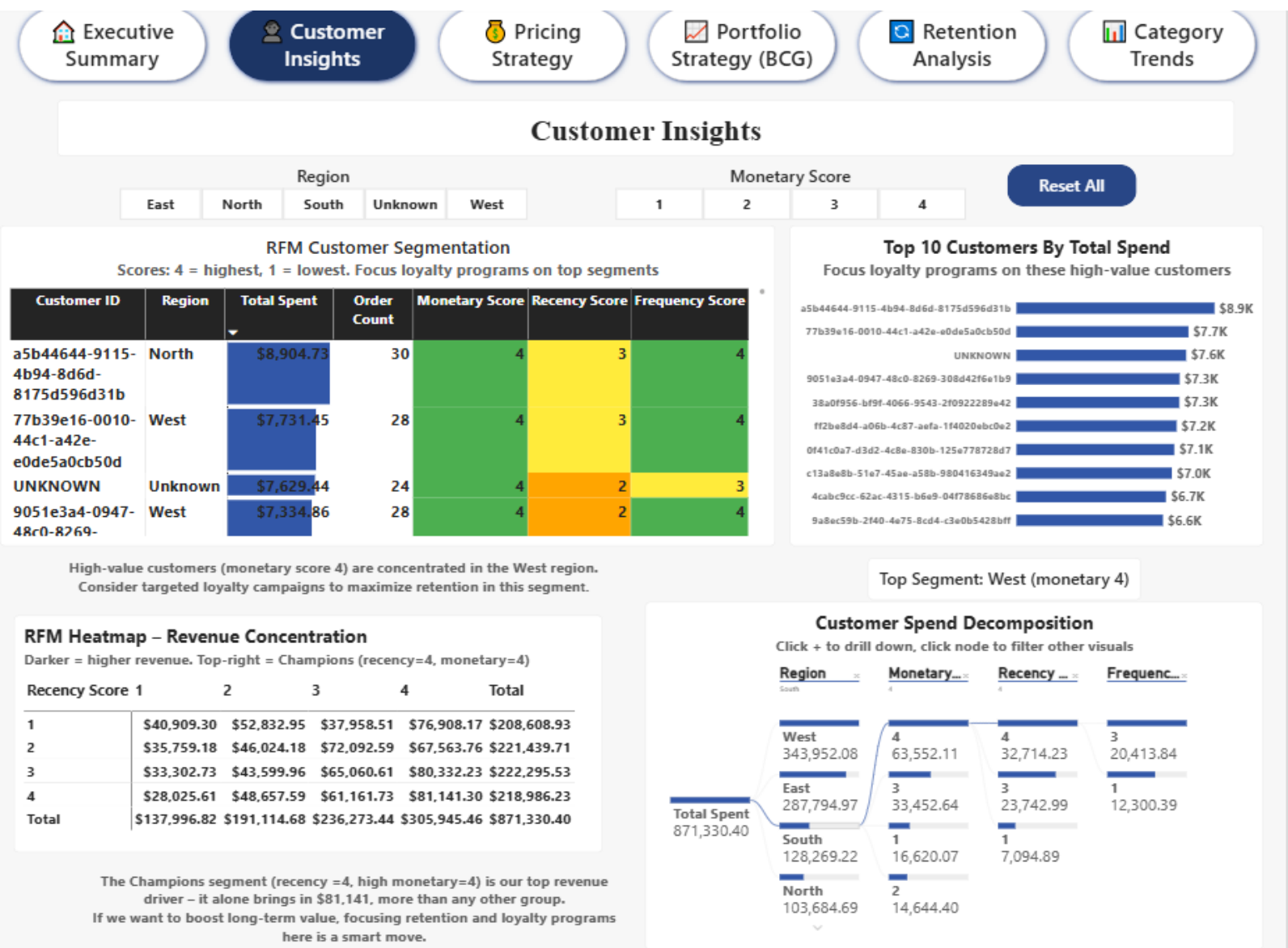
- Steady growth to \$871K — consistent upward trend
- Forecast: Projected Revenue: \$871.33K
- Forecast: Projected Profit: \$474.53K

Top Category: Grinders & Brewers — 58% of total revenue

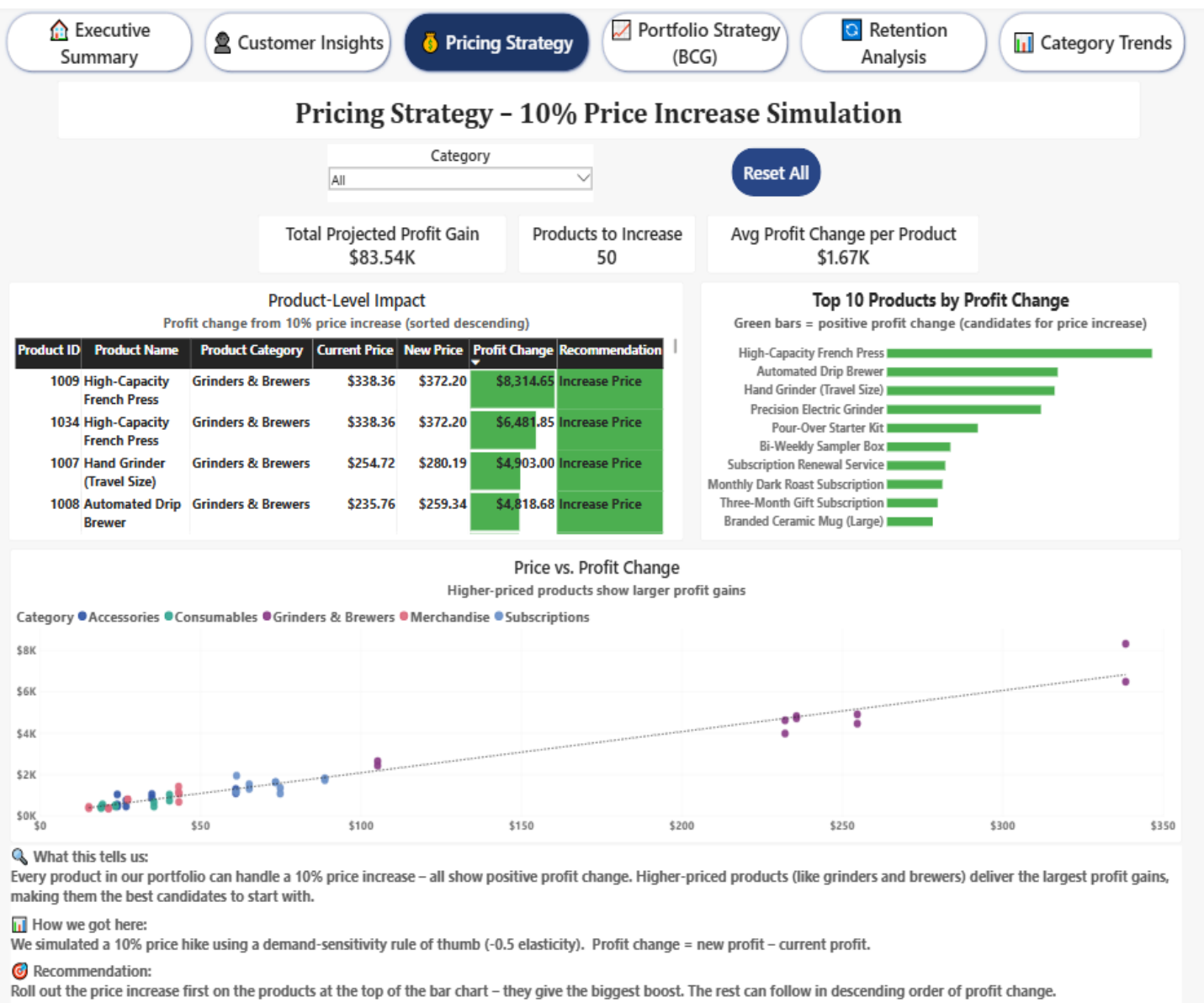
Key Insight:

"Grinders & Brewers dominate revenue & profit — clear Cash Cow".

Customer Insights — Who Are Our Best Customers?



Pricing Strategy — 10% Price Increase Simulation



Impact Summary:

- Total Projected Profit Gain: \$83.54K
- Products to Increase: 50
- Avg Profit Change per Product: \$1.67K

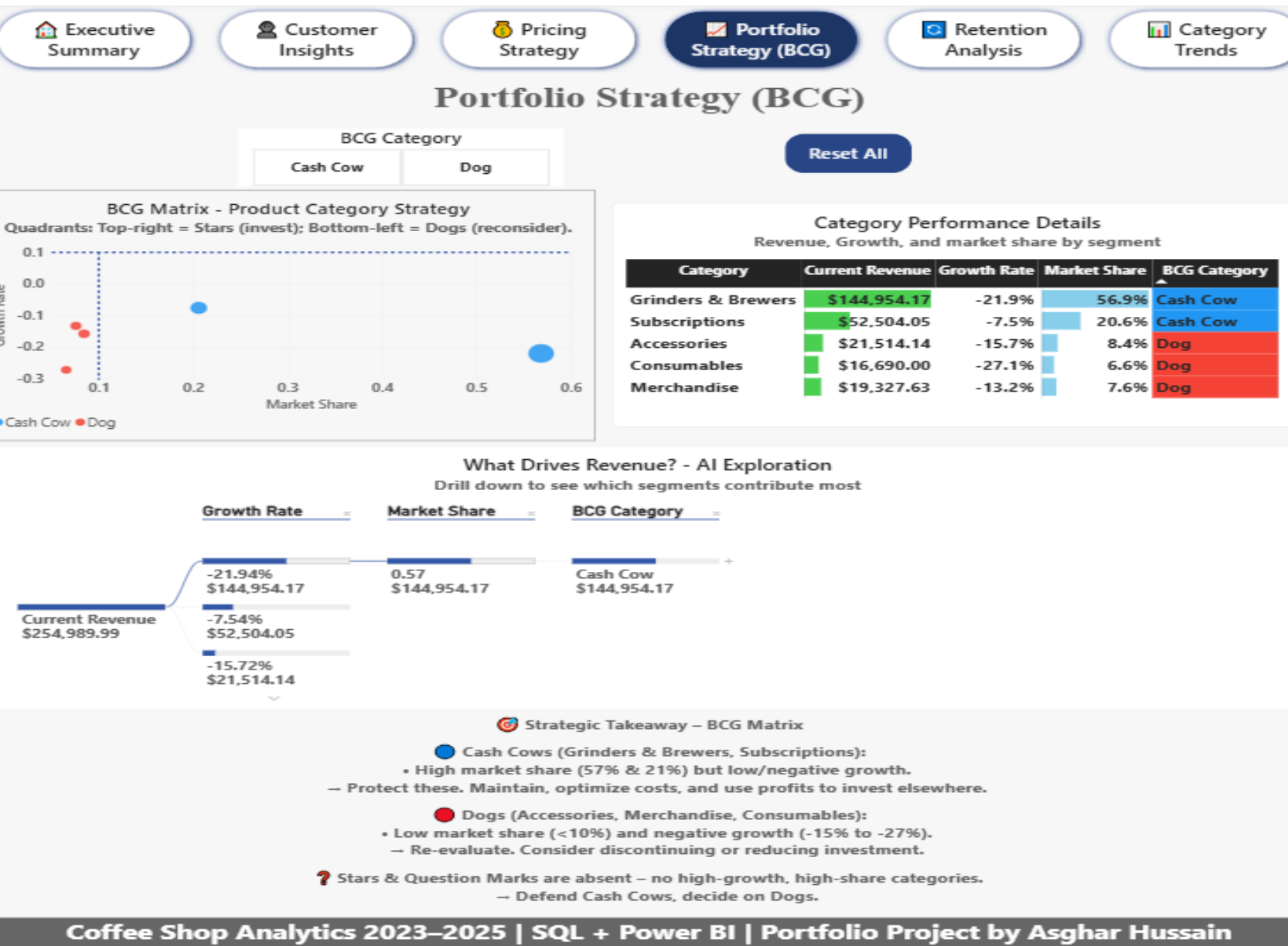
Top 3 Products by Profit Change:

1. High-Capacity French Press — \$8,314.65
2. High-Capacity French Press (variant) — \$6,481.85
3. Hand Grinder (Travel Size) — \$4,903.00

Key Insight:

"Higher-priced products (Grinders & Brewers) deliver the largest profit gains."

Portfolio Strategy — BCG Matrix



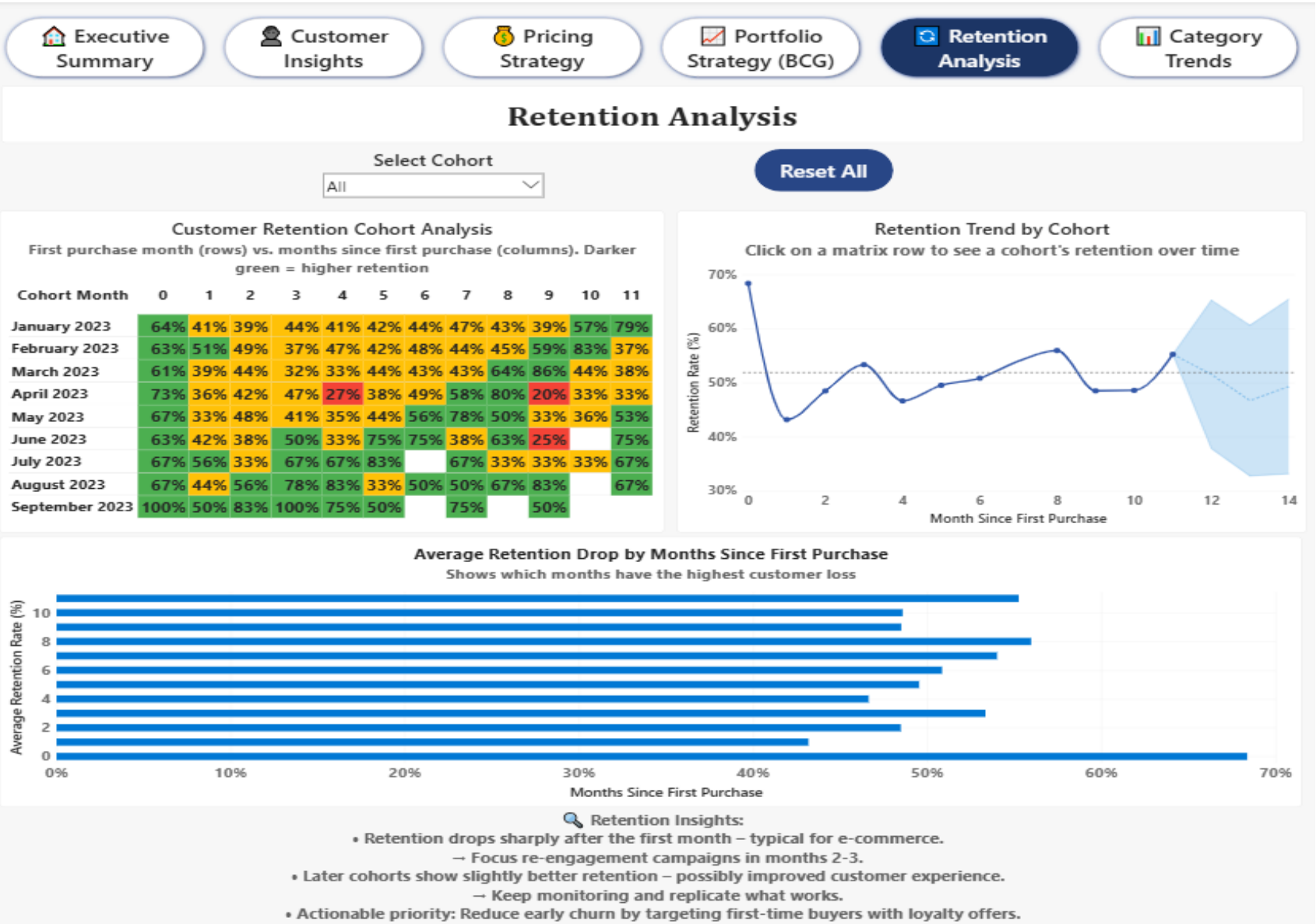
Category Strategy:

- Cash Cow: Grinders & Brewers — Protect
- Question Marks: Subscriptions — Invest carefully
- Dogs: Accessories, Merchandise, Consumables — Re-evaluate

Key Insight:

"Protect Cash Cow. Re-evaluate Dogs for discontinuation."

Retention Analysis — Do Customers Come Back?



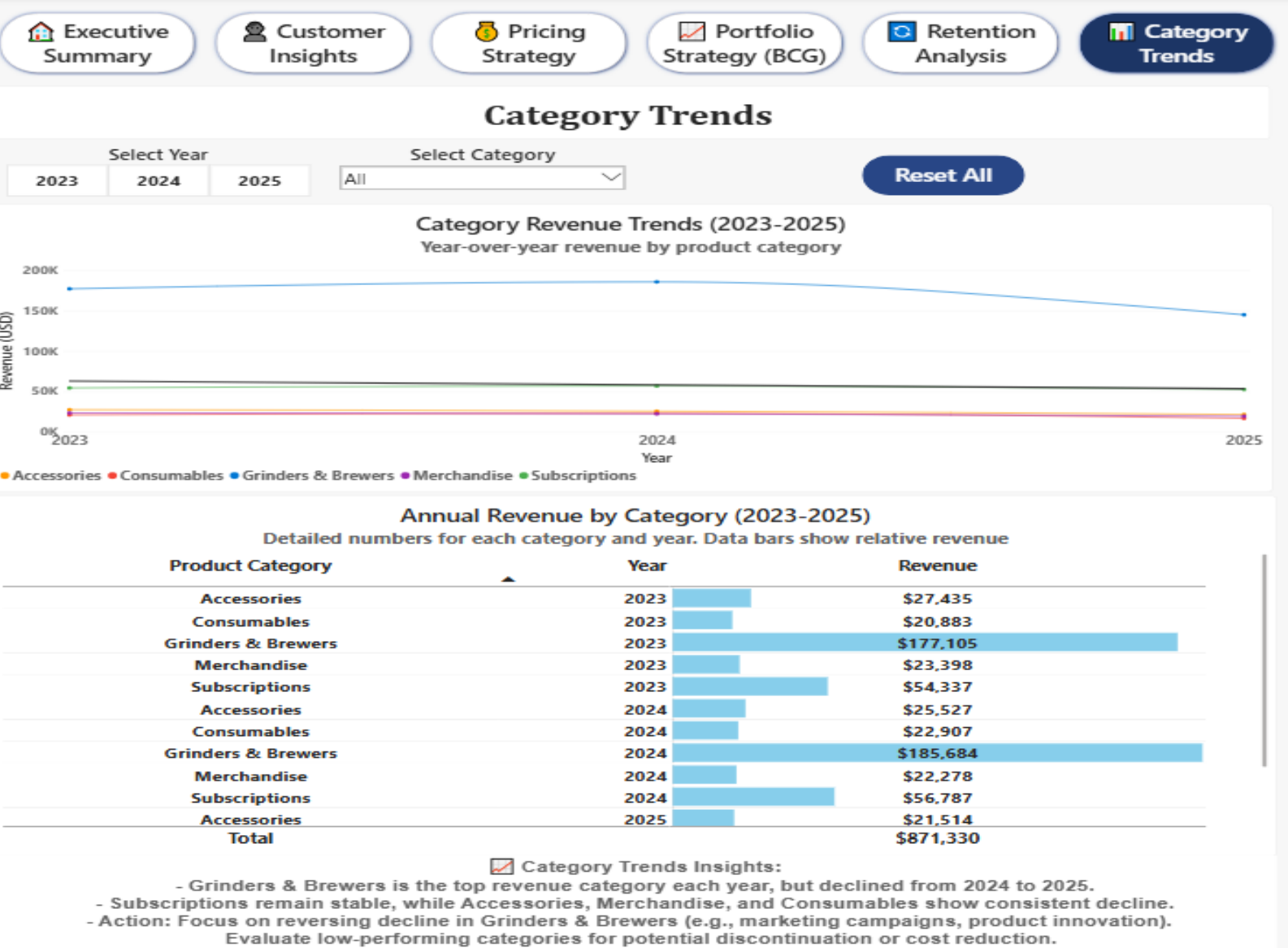
Retention Findings:

- Month 1: 100%
- Month 2: ~45%
- Month 3: ~33%
- Drop: 55% in month 2
- Later cohorts (2024-2025) show slightly better retention

Key Insight:

"Retention drops sharply after the first month — focus re-engagement campaigns on months 2–3."

Category Trends — YoY Revenue by Category



Category Performance:

- Grinders & Brewers: Top revenue (declining)
- Subscriptions: Stable
- Accessories, Merchandise, Consumables: Declining

Key Insight:

"Grinders & Brewers declining since 2024 — investigate causes."

Technical Architecture — How I Built It

Phase 1: PostgreSQL — Data Loading & Advanced Views

- Normalized tables (customers, products, orders)
- Used COPY commands to import 3 years of CSV data
- Built 5 advanced views using CTEs, window functions, self-joins:
 - RFM segmentation
 - Price simulation (10% hike)
 - BCG matrix (Market Share vs Growth Rate)
 - Cohort retention analysis
 - Category trends (YoY revenue)

Phase 2: Power BI — Dashboard & UX Design

- 6 interactive pages with consistent theme
- Conditional formatting (data bars, colors)
- DAX measures: KPIs, YoY growth, what-if parameters
- Tooltips, slicers, reset buttons

Phase 3: AI Integration — Smart Assistant

- Claude AI: SQL optimization, DAX generation, documentation
- DeepSeek: Troubleshooting & code refinement
- Reduced development time by ~40%

 Thank You!

Hi! I'm a data analyst passionate about turning messy data into clear, actionable business insights. This project reflects my approach: business-first, data-driven, and human-centered.

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If you found this project useful, please star the repository!

Built with  and curiosity.